QC Foundations [1/4]

Quantum Computing — CEP

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Where We Are

Previous Week (Oct 17)

- Complex numbers, $\mathbb{C}$
- $\mathbb{C}^2$ column vectors
- Inner product $\langle u, v \rangle$, norms
- Orthonormal bases
- Projection onto a subspace

Today we add

- **Physical meaning**: what lives in $\mathbb{C}^2$
- Dirac ket $|\psi\rangle$ — a compact name
- Geometry: the Bloch sphere
- Operations: gates as rotations
- Measurement: the Born rule

Today's promise

No mathematics today beyond what has been covered last week. We will see what the *physics* does with it.

Ket Notation — Just a Shorthand

$$|\psi\rangle \;=\; \alpha\,|0\rangle + \beta\,|1\rangle \;=\; \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$$

- $|\psi\rangle$ is a name for a column vector in $\mathbb{C}^2$
- The label inside $|\cdot\rangle$ is just a tag
- $|0\rangle$, $|+\rangle$, $|\Phi^+\rangle$ — all column vectors
- Dirac invented this to make multi-qubit expressions readable

Key examples

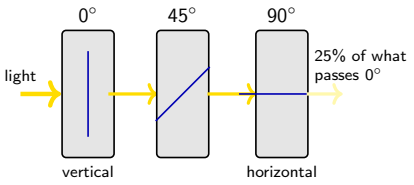
$$|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

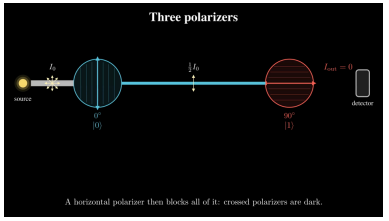
We use $|\cdot\rangle$ throughout the course. Bras $\langle\cdot|$, inner products, and outer products are formalised in a later Session.

A Puzzle — Three Polarising Filters

Two crossed filters (0° and 90°) block all light. Insert a 45° filter between them: light passes. How?



Real-world demo (2 min)



Animation: the paradox (25 s)

*Not a trick: quantum mechanics in your hands. *calculate* the 25%.*

Hold this question. Later today you will

Polarisation IS a Qubit

Physical state	Notation
Vertical $\lvert V\rangle$	$\lvert 0\rangle$
Horizontal $\lvert H\rangle$	$\lvert 1\rangle$
45° diagonal	$\lvert +\rangle = \dfrac{\lvert 0\rangle + \lvert 1\rangle}{\sqrt{2}}$

Not an analogy

Photon polarisation is one of the *physical realisations* of a qubit — not a metaphor for one.

Others: electron spin ↑ / ↓, superconducting circuit energy levels, trapped-ion internal states, NV centres in diamond.

The Classical Bit

- State $\in \{0, 1\}$ — exactly two isolated points
- Operations: deterministic maps $\{0,1\} \to \{0,1\}$
- One measurement always returns exactly one bit
- **State space:** finite, discrete

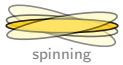
Everything a classical computer does is ultimately sequences of bits and bit operations.



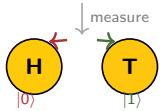
state space: 2 points

Superposition — Two Ways to See It

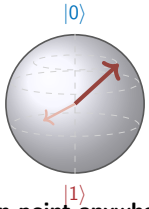
The spinning coin



neither H nor T



The compass needle



can point anywhere

infinite state space

Superposition is not "being in two places at once". It is a definite, specific state — just not a classical one.

The Qubit's Home — $\mathbb{C}^2$

A qubit state is a **unit vector** in $\mathbb{C}^2$:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad \alpha, \beta \in \mathbb{C}$$

Physical constraint (Born rule)

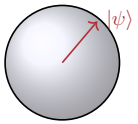
$$|\alpha|^2 + |\beta|^2 = 1$$

Probabilities of all outcomes must sum to 1.

Computational basis (ONB):

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Every qubit state: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$.

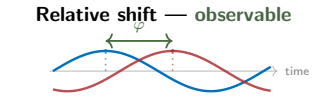


qubit: sphere ($\mathbb{C}^2$)

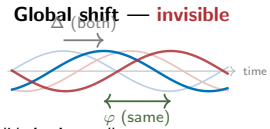


classical: 2 points

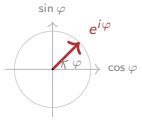
Phase — A Classical Warm-up



the two waves **differ** ⇒ interference changes



slide **both** equally ⇒ same pattern

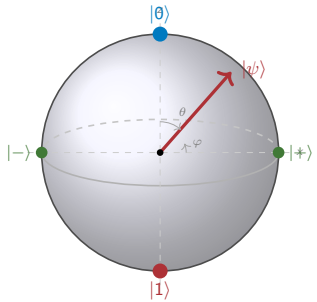


phase = an angle

- Phase is an angle $\varphi \in [0, 2\pi)$ on a circle: the phasor $e^{i\varphi} = \cos\varphi + i\sin\varphi$.
- Only the **relative** phase of two waves is observable ($\varphi = \pi$ cancels — noise-cancelling headphones; $\varphi = 0$ reinforces).
- A **global** shift moves both together — measurements cannot see it.

Phase is an angle, and only relative phase matters.

The Bloch Sphere



Every qubit state is a point on the unit sphere:

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}\,|1\rangle$$

- $\theta \in [0, \pi]$ — **latitude**
- $\varphi \in [0, 2\pi)$ — **longitude**
- North pole: $|0\rangle$
- South pole: $|1\rangle$
- Equator: equal superposition

$\mathbb{C}^2$ has 4 real dimensions. Normalisation removes 1. Global phase removes 1 more. Two dimensions remain — a sphere.

Why Two Angles Suffice

A general qubit is two complex numbers:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad \alpha, \beta \in \mathbb{C} \quad \Longrightarrow \quad \textbf{4} \text{ real parameters.}$$

Two physical facts cut four down to two.

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$$|\alpha|^2 + |\beta|^2 = 1$$

One constraint. **4** → **3**.

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2. Global phase is invisible

Multiply by $e^{-i \arg \alpha}$ — no measurable change — so take α **real**, ≥ 0. $\mathbf{3} \to \mathbf{2}$.

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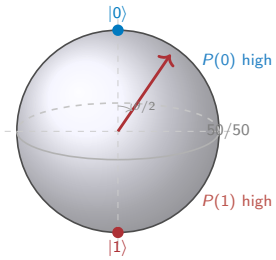
Multiply by $e^{-i \arg \alpha}$ — no measurable change — so take α **real**, ≥ 0. $\mathbf{3} \to \mathbf{2}$.

With α real in $[0, 1]$, set $\alpha = \cos\frac{\theta}{2}$ (so $\theta \in [0, \pi]$); normalisation forces $|\beta| = \sin\frac{\theta}{2}$, and the *one* phase left over rides on β:

$$\boxed{|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\varphi} \sin\tfrac{\theta}{2}\,|1\rangle}$$

θ = latitude (sets the measurement odds), φ = longitude (the relative phase from the last slide). Two angles = a point on a sphere.

Latitude Controls Probability



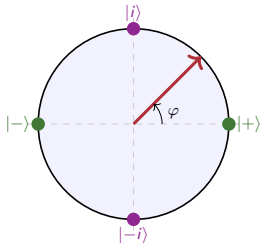
$$P(0) = |\alpha|^2 = \cos^2\tfrac{\theta}{2}$$

$$P(1) = |\beta|^2 = \sin^2\tfrac{\theta}{2}$$

State	θ	$P(0)$
$\|0\rangle$	0	1
$\|+\rangle$	$\pi/2$	1/2
$\|1\rangle$	π	0

Closer to the north pole: higher $P(0)$.
On the equator: exactly 50/50.

Longitude Controls Phase



view from above (equatorial plane)

The coefficient $\beta = e^{i\varphi} \sin\frac{\theta}{2}$ where

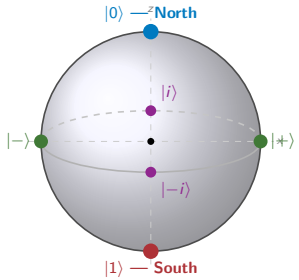
$$e^{i\varphi} = \cos\varphi + i \sin\varphi$$

Longitude is a **rotation in the complex plane**.

State	φ	β
$\|+\rangle$	0	$\frac{1}{\sqrt{2}}$
$\|i\rangle$	$\pi/2$	$\frac{i}{\sqrt{2}}$
$\|-\rangle$	π	$\frac{-1}{\sqrt{2}}$
$\|-i\rangle$	$3\pi/2$	$\frac{-i}{\sqrt{2}}$

Same latitude = same measurement odds. Different longitude = different interference behaviour.

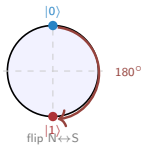
Six Key States



State	(θ, φ)	Vector
$\|0\rangle$	$(0, -)$	$\begin{pmatrix}1\\0\end{pmatrix}$
$\|1\rangle$	$(\pi, -)$	$\begin{pmatrix}0\\1\end{pmatrix}$
$\|+\rangle$	$(\pi/2, 0)$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$
$\|-\rangle$	$(\pi/2, \pi)$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$
$\|i\rangle$	$(\pi/2, \pi/2)$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}$

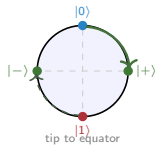
Quantum Gates are Rotations of the Sphere

X gate



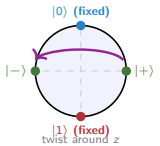
$|0\rangle \leftrightarrow |1\rangle$

H gate



$|0\rangle \to |+\rangle$, $|1\rangle \to |-\rangle$

Z gate



$|+\rangle \to |-\rangle$; poles fixed

A Gate is a Linear Function $\mathbb{C}^2 \to \mathbb{C}^2$

A gate U takes a state in, gives a state out:

$$U|\psi\rangle = \begin{pmatrix}u_{00} & u_{01}\\ u_{10} & u_{11}\end{pmatrix}\begin{pmatrix}\alpha\\\beta\end{pmatrix} = \begin{pmatrix}u_{00}\alpha + u_{01}\beta\\ u_{10}\alpha + u_{11}\beta\end{pmatrix}$$

Key requirements:

- Maps unit vectors to unit vectors (**unitary**)
- Always reversible
- **Linear:** $U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle$

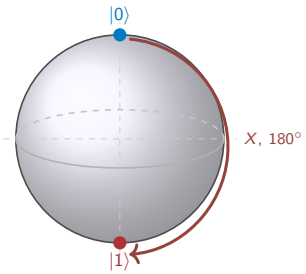
The X gate

$$X = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

$$X|0\rangle = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} = |1\rangle \quad 3$$

Linear transformations are formalised in Session 3/4. Today we use them visually.

The X Gate — Quantum NOT



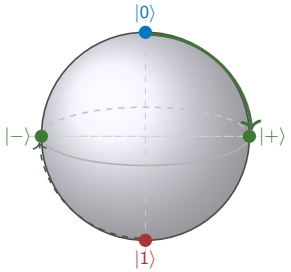
$$X = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

Input	Output	
$\|0\rangle$	$\|1\rangle$	flip
$\|1\rangle$	$\|0\rangle$	flip
$\|+\rangle$	$\|+\rangle$	*fixed!*
$\|-\rangle$	$-\|-\rangle$	phase sign only

Geometrically

180° rotation around the x-axis. Swaps north and south poles.

The H Gate — Superposition Machine



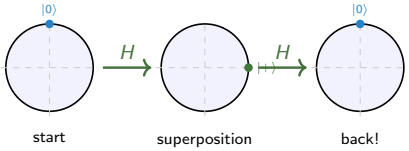
$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Input	Output
$\|0\rangle$	$\|+\rangle = \frac{\|0\rangle+\|1\rangle}{\sqrt{2}}$
$\|1\rangle$	$\|-\rangle = \frac{\|0\rangle-\|1\rangle}{\sqrt{2}}$
$\|+\rangle$	$\|0\rangle$
$\|-\rangle$	$\|1\rangle$

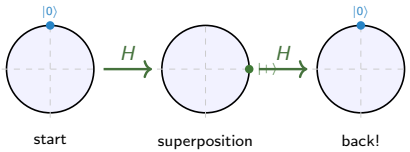
Geometrically

180° rotation around the axis halfway between x and z. Takes poles to equator and back.

Apply H Twice — Interference in Action



Apply H Twice — Interference in Action



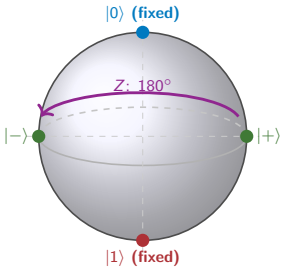
Why?

$$H|+\rangle = \frac{H|0\rangle+H|1\rangle}{\sqrt{2}} = \frac{|+\rangle+|-\rangle}{\sqrt{2}} = \frac{(|0\rangle+|1\rangle)+(|0\rangle-|1\rangle)}{2} = |0\rangle$$

Interference

The $|1\rangle$ amplitudes **cancel** (destructive interference). The $|0\rangle$ amplitudes **reinforce** (constructive interference).

The Z Gate — Phase Twist



$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Input	Output	
$\|0\rangle$	$\|0\rangle$	*(fixed)*
$\|1\rangle$	$-\|1\rangle$	*(phase flip)*
$\|+\rangle$	$\|-\rangle$	*(twist)*
$\|-\rangle$	$\|+\rangle$	*(twist back)*

Geometrically

180° rotation around the z-axis.
Poles fixed; equator swaps $|+\rangle \leftrightarrow |-\rangle$.

Poll — Where Does $HZH\,|0\rangle$ Land?

Question

Apply H, then Z, then H to $|0\rangle$. What is the final state?

A) $|0\rangle$ **B)** $|1\rangle$ **C)** $|+\rangle$ **D)** $|-\rangle$

Poll — Where Does $HZH\,|0\rangle$ Land?

Question

Apply H, then Z, then H to $|0\rangle$. What is the final state?

A) $|0\rangle$ **B)** $|1\rangle$ **C)** $|+\rangle$ **D)** $|-\rangle$

Step by step:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle$$

Poll — Where Does $HZH\,|0\rangle$ Land?

Question

Apply H, then Z, then H to $|0\rangle$. What is the final state?

A) $|0\rangle$ **B)** $|1\rangle$ **C)** $|+\rangle$ **D)** $|-\rangle$

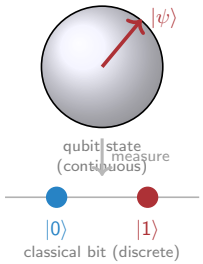
Step by step:

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle$$

Answer: B — $|1\rangle$

HZH flips a qubit just like X does. In fact, $HZH = X$.
This is Hadamard *conjugation*: it converts a z-rotation into an x-rotation. You will see this pattern throughout the course.

Measurement — The Shadow Analogy



- A 3D state is forced onto a **1-bit answer**
- Like a 3D object casting a 2D shadow
- Outcome is **random**, governed by latitude
- Measurement **destroys** the superposition

No going back

After measurement: state has collapsed to $|0\rangle$ or $|1\rangle$. Measuring again gives the *same* outcome.

The state before measurement was not "secretly $|0\rangle$ or $|1\rangle$". The randomness is fundamental — not ignorance.

The Born Rule

For $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$:

$$P(\text{outcome } 0) = |\alpha|^2, \qquad P(\text{outcome } 1) = |\beta|^2$$

On the Bloch sphere

$$P(0) = \cos^2\tfrac{\theta}{2}, \qquad P(1) = \sin^2\tfrac{\theta}{2}$$

Latitude controls the odds.

Connection to inner product:

$$P(0) = |\,\langle 0|\psi\rangle\,|^2 = |\alpha|^2$$

(Full derivation in Session 3/4.)

State	$P(0)$	$P(1)$
$\|0\rangle$	1	0
$\|1\rangle$	0	1
$\|+\rangle$	1/2	1/2
$\frac{1}{2}\|0\rangle + \frac{\sqrt{3}}{2}\|1\rangle$	1/4	3/4

Our filter

puzzle: 25% $= \frac{1}{2} \times \frac{1}{2}$ is the Born rule applied twice. We watch it happen in two frames' time.

Worked Example — Measuring $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$$

What are the measurement outcomes?

Worked Example — Measuring $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$$

What are the measurement outcomes?

Step 1: $\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$

Worked Example — Measuring $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$$

What are the measurement outcomes?

Step 1: $\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$

Step 2: $P(0) = |\alpha|^2 = \frac{1}{2}, \quad P(1) = |\beta|^2 = \frac{1}{2}$

Worked Example — Measuring $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

What are the measurement outcomes?

Step 1: $\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$

Step 2: $P(0) = |\alpha|^2 = \frac{1}{2}, \quad P(1) = |\beta|^2 = \frac{1}{2}$

Step 3: 50% chance of outcome 0; 50% chance of outcome 1.

Worked Example — Measuring $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

What are the measurement outcomes?

Step 1: $\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$

Step 2: $P(0) = |\alpha|^2 = \frac{1}{2}, \quad P(1) = |\beta|^2 = \frac{1}{2}$

Step 3: 50% chance of outcome 0; 50% chance of outcome 1.

Step 4: After outcome 0: state **collapses** to $|0\rangle$.

Connection to the filter

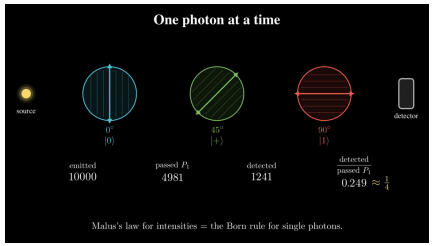
This is exactly why the 45° filter transmitted 50% — photons in $|+\rangle$ each face $P(1) = \frac{1}{2}$ at the 90° filter.

Trap: $|+\rangle \neq$ a coin flip

$|+\rangle$ gives 50/50 outcomes *but* is not classical uncertainty. It is a definite quantum state. The difference shows up in interference — apply H and it returns to $|0\rangle$.

The Filter Puzzle, Solved — Watch It

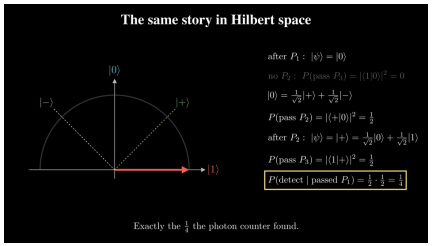
Act 3: one photon at a time



▸ Play (26 s)

Each photon passes or is absorbed *at random*; only the probability is fixed. The counter settles at $\frac{1}{4}$.

Act 4: the same story as shadows



▸ Play (36 s)

This morning's shadow analogy, applied twice: $|0\rangle \to |+\rangle \to |1\rangle$, probability $\frac{1}{2} \cdot \frac{1}{2}$.

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$$P(\text{detected} \mid \text{passed } 0°) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4} = 25\%$$

Key insight

Each filter is a **measurement** in its own basis. A photon that passes the 45° filter *is now* $|+\rangle$: the record of "vertical" is erased, so the 90° filter gets a fresh 50/50 chance. Without the middle filter, $|\langle 1|0\rangle|^2 = 0$ and nothing passes.

Relative Phase is Observable

Consider $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Same measurement probabilities: both give $P(0) = P(1) = \frac{1}{2}$.

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$$H|+\rangle = |0\rangle\,, \qquad H|-\rangle = |1\rangle$$

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Relative phase

The sign difference between $|+\rangle$ and $|-\rangle$ is a **relative phase** — physically real, observable through interference.

Global phase is NOT observable

$e^{i\theta}|\psi\rangle$ and $|\psi\rangle$ give identical measurement results for every experiment.
Only the *relative* phase between $|0\rangle$ and $|1\rangle$ components matters.

This is longitude on the Bloch sphere. Same latitude = same probabilities. Different longitude = different interference.

Polariser ↔ Qubit Dictionary

Polarizer ↔ qubit dictionary

Optics	Qubit
Polarization at angle θ	$\|\theta\rangle = \cos\theta\|0\rangle + \sin\theta\|1\rangle$
Polarizer at angle θ	Measurement in basis $\{\|\theta\rangle, \|\theta{+}90°\rangle\}$
Malus's law, $\cos^2\Delta\theta$	Born rule, $\|\langle\theta\|\psi\rangle\|^2$
Transmitted light is re-polarized	Post-measurement state (collapse)
Absorbed photon	The other outcome, $\|\theta{+}90°\rangle$

Caveat: on the Bloch sphere every angle *doubles*. Orthogonal polarizations ($0°$, $90°$) are antipodal poles; $45°$ lands on the equator as $|+\rangle$. Linear polarizations cover only one great circle (real amplitudes); circular polarizations supply the complex phases, $|\pm i\rangle$.

▸ Play Act 5 (15 s)

One line

A polariser is a measurement; Malus's law is the Born rule.

Caveat: angles double

$0°$ and $90°$ are *orthogonal* polarisations, so they sit at opposite poles of the Bloch sphere. $45°$ lands on the equator as $|+\rangle$.

Break

10 minutes

Two Qubits — The 4-State Space

Two qubits live in $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

with $\sum_{ij} |\alpha_{ij}|^2 = 1$.

Basis state	q_1	q_0
$\|00\rangle$	0	0
$\|01\rangle$	0	1
$\|10\rangle$	1	0
$\|11\rangle$	1	1

Exponential scaling

Qubits	**Amplitudes**
1	2
2	4
10	1,024
50	$\sim 10^{15}$
300	$> 10^{90}$

10^{90} exceeds the number of atoms in the observable universe. A classical computer cannot store this state.

Tensor Product — Preview

Two separate qubits $|a\rangle$ and $|b\rangle$ combine as:

$$|a\rangle \otimes |b\rangle \equiv |ab\rangle$$

Product state (separable)

Each qubit has its own state. Can always be written $|a\rangle \otimes |b\rangle$.

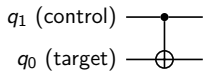
Example: $|0\rangle \otimes |1\rangle = |01\rangle$

Entangled state

Cannot be written as $|a\rangle \otimes |b\rangle$.
The qubits are *correlated*, not independent.

Amit formalises the Kronecker product on Sunday Oct 25. Today we use $\otimes$ as notation only.

The CNOT Gate



Rule: if $q_1 = 1$, flip q_0; otherwise do nothing.

Input		**Output**	
q_1	q_0	q_1	q_0
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

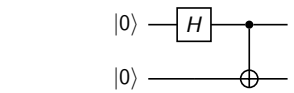
In superposition

CNOT acts on *all* basis states simultaneously:

$$\text{CNOT}\,\frac{|00\rangle + |10\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The right-hand side cannot be written as $|a\rangle \otimes |b\rangle$ for any single-qubit states — it is entangled.

Building a Bell Pair

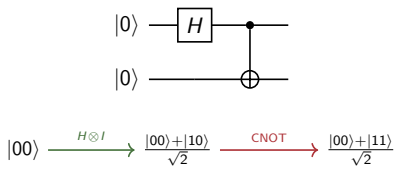


$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle+|10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

Step 1: H on q_1 (top):

$$|00\rangle \to \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Building a Bell Pair



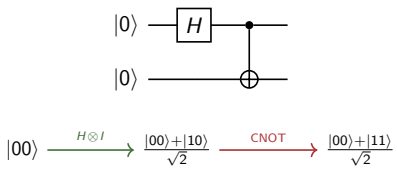
Step 1: H on q_1 (top):

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Step 2: CNOT:

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Building a Bell Pair



Step 1: H on q_1 (top):

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Step 2: CNOT:

$$\frac{|00\rangle+|10\rangle}{\sqrt{2}} \to \frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

Result: $|\Phi^+\rangle$

$$|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

The Bell State — Spooky Correlations

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- Outcome 0: q_1 **instantly becomes** $|0\rangle$
- Outcome 1: q_1 **instantly becomes** $|1\rangle$
- Perfectly correlated, no matter how far apart

Cannot be separated

$|\Phi^+\rangle \neq |a\rangle \otimes |b\rangle$ for any $|a\rangle$, $|b\rangle$.
This is **entanglement**.

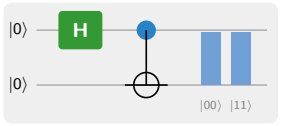
Not telepathy

The correlations are real but cannot transmit information faster than light. (Proof in Session 4/4.)

Einstein: "spooky action at a distance." Bell (1964): correlations exceed any classical explanation. Aspect (Nobel 2022): experimentally confirmed.

From Quirk to Qiskit

Quirk (browser, drag-and-drop)



Open Quirk (algassert.com/quirk)

Qiskit (Python, on merlin)

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
```

Same circuit. You will write Qiskit in every lab and capstone. Today we use Quirk to *watch the states move.*

Both produce $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

Sandbox — 10 Minutes

Open Quirk (link in chat)

Puzzle 1

Start from $|0\rangle$. Reach $|1\rangle$ with **100% certainty** using only H and Z gates (no X).

Puzzle 2

Create the Bell state $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ instead of $|00\rangle + |11\rangle$. Hint: one extra gate, on the bottom wire.

Puzzle 3 (stretch)

What single-gate circuit returns $|+\rangle$ back to $|0\rangle$? Explain why in terms of interference.

What's Next

This afternoon — Session 2/4

Your first quantum speedup

The Deutsch-Jozsa algorithm:

- Black-box oracles
- Quantum parallelism
- Phase kickback
- Interference wins over classical queries

No classical algorithm can match it in one query.

Sunday Oct 25 (Amit)

Tensor products: formal Kronecker product, bilinearity, multi-qubit operators

Saturday Oct 31 — Session 3/4

Formalising today: Dirac bra-ket algebra, gates as outer products and matrices, tensor products, and the algebraic proof of Deutsch–Jozsa

Reading

Mermin (start here)

- §1.1–1.3: qubits and state space
- §1.4–1.5: unitary operations
- §1.6: measurement

Wong (free online)

- Ch. 3: qubits and gates
- Ch. 4: measurement

Nielsen & Chuang (deeper)

§1.2, §2.1.1–2.1.4 for the formal state space; §2.2.1, §2.2.3 for single-qubit gates

You do not need to read everything before Session 2/4 this afternoon. Read at least Mermin §1.1–1.3 and try to rederive the Born rule calculation for the 45° polarising filter.